TECHNICAL SPECIFICATIONS

SECTION 26 13 26

MEDIUM VOLTAGE (MV) SWITCHGEAR

Project: Meridian Data Centre Campus, Phase 1
Navi Mumbai, India
Revision: 2024 (December 12, 2024)

PART 1 - GENERAL

1.1 Summary

This section specifies metal-clad, air-insulated switchgear for primary distribution at 11kV nominal utility voltage.

1.2 References

- IEC 62271-200 - AC metal-clad switchgear and controlgear for rated voltages above 1kV
- IEC 62271-1 - High-voltage switchgear and controlgear common specifications
- IS 3427 - AC metal-clad switchgear and controlgear for voltages above 1kV

1.3 Submittal Requirements

A. Contractor shall submit the following for Engineer's approval prior to equipment procurement:

1. Product Data: Technical datasheets detailing all electrical and physical parameters, performance curves, and compliance.
2. Shop Drawings: Dimensional layouts showing clearance requirements, cabling paths, and weights.
3. Compliance Matrix: Clause-by-clause confirmation of conformance. Any deviation must be explicitly noted.
4. Factory Test Certificates and Type Test Certifications.

1.4 Quality Assurance

A. Equipment shall be manufactured by a firm specializing in the equipment class with a minimum of twenty (25) years of continuous commercial manufacturing history.

B. As specified in Section 26 05 00 Part 1.4.3.B, all medium voltage switchgear and high-power electrical components must be supplied with certified test certificates from an internationally recognized independent laboratory (ASTA, KEMA, or equivalent) confirming compliance with IEC standards.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

A. Subject to compliance with technical requirements, approved manufacturers are limited to the following:

2.2 Performance and Design Requirements

rated_voltage: Rated maximum voltage must be >= 12 kV.

bus_current: Continuous main busbar current rating must be >= 2000 A.

short_circuit: Short-time withstand rating must be >= 31.5 kA for 3 seconds.

protection_rating: Enclosure degree of protection must be IP4X (outer cover) and IP2X (internal partitions).

arc_classification: Internal Arc Classification (IAC) must be AFLR, rated for 31.5 kA for 1.0 second.

> **Injected Contract Defects (Internal Quality Control Audit Targets):**
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> - **DEF-003:** Common Electrical Section 26 05 00 Part 1.4.3.B mandates IEC test certificates for all medium voltage gear, but Section 26 13 26 Part 1.2.1 references UL standards only for certification, causing a conflict. (Location: 26 05 00 Part 1.4.3.B vs 26 13 26 Part 1.2.1)

PART 3 - EXECUTION

3.1 Installation and Examination

A. Verify that structural concrete floor slab is fully cured. Verify structural load limit of 2,800 kg/m^2 is not exceeded in UPS or battery rooms.

B. Maintain a minimum clear work clearance of 1000 mm in front of all enclosures, and 1200 mm above, in accordance with local Indian Electricity Rules and IEC standards.

3.2 Field Quality Control and Site Commissioning

A. Startup services shall be performed by factory-authorized technicians.

B. Commissioning tests must be scheduled and carried out in accordance with the project's overall Cx Test Register. Every testing step must map back to a specific specification clause.

dielectric_tests: Conduct power frequency withstand and lightning impulse voltage testing.

contact_resistance: Measure busbar joint contact resistances using micro-ohmmeter.